Problem Set - 6

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1 Mechanics

1.1 Balancing a Ruler

A ruler of length L and mass m is placed on two fingers by it's two ends. The static and kinetic coefficients of friction between the fingers and the ruler is given as μ_s and μ_k ($\mu_s > \mu_k$), respectively. The person holding the ruler starts bringing his fingers closer to each other slowly. What will be the total work that is converted to heat by the time two fingers meet? (3. Aşama, 2024)

1.2 Kapitza's Pendulum

As shown in the figure, a rigid uniform rod of length l and mass m rotates freely around a horizontal axis O, to which it is attached at one end. The position of the rod in the plane perpendicular to the axis of rotation is described by the angle $\theta \in (-\pi, \pi)$ between the rod and the vertical downward direction. Angle θ is counted counterclockwise. The acceleration of gravity is equal to g. Neglect air resistance when solving the problem. Axis of rotation O oscillates vertically with small amplitude $A \ll l$ and high frequency $\omega \gg \frac{g}{l}$ in law:

$$z(t) = A \cos \omega t$$

Let us consider the motion of the rod in a non-inertial frame of reference associated with the axis O.

i) Write the equation of dynamics for the dependence of the angle $\theta(t)$ on time.

Angle dependence $\theta(t)$ from time can be approximately written as the sum $\varphi(t)$ and $\delta(t)$, where $\varphi(t)$ represents a slowly changing average component of this dependence, and $\delta(t)$ represents harmonic oscillations with a frequency ω. $\varphi(t)$ changes slowly enough that it can be considered approximately constant on time scales of the order of the period $\delta(t)$. The influence of the high-frequency component on the motion of the rod can be investigated by averaging the dynamics equation over the period.

ii) Derive an equation that satisfies $\varphi(t)$.

The equation obtained in the previous paragraph can be interpreted as the equation of motion of the rod in an effective external potential $V_{\text{eff}}(\varphi)$.

iii) Considerind that $V_{\text{eff}}(\varphi = 0) = 0$, get the expression for $V_{\text{eff}}(\varphi)$.

iv) Determine the equilibrium positions φ_0 of the rod in the field $V_{\text{eff}}(\varphi)$. Investigate these equilibrium positions for stability. Find the frequency of small oscillations relative to stable equilibrium positions. Consider all possible cases.

Let at the initial moment of time the rod be directed vertically downwards ($\theta(t = 0) = 0$) and moves with angular velocity $\dot{\varphi}(t = 0) = \omega_0 > 0$. In order for the rod to make a full revolution, its angular velocity must be greater than a certain critical value ω_c.

v) Find ω_c and the maximum angle $\varphi_{\max}$, to which the rod can rise in case $\omega_0 < \omega_c$. Consider all possible cases. (CN23)

2 Electromagnetism

2.1 Magnetostatics

In 3-D space, a permeable medium covers the region $x > 0$, while the rest of the space is vacuum. The medium's relative magnetic permeability is $\mu_r > 1$. A magnetic dipole with dipole moment m is placed a distance d away from the permeable medium, at position $(-d, 0, 0)$. The dipole is pointed towards the $+x$ direction. Treat the dipole as ideal (point-sized). (OPhO 2020, Invitational, P10)

(a) Find the force required to keep the dipole in place.

(b) How much work does it take to slowly pull the dipole from its original position to infinity (at $x = -\infty$)?

(c) How much work does it take to slowly rotate the dipole from its original orientation to one that makes an angle θ with the $+x$-axis?

After the dipole is rotated an angle θ, a superconducting ring with radius R and self-inductance L is brought in from infinity (with initially no current). It is placed so that the dipole is located at its center and its axis is the x-axis. Assume that $R \gg d$.

(d) Find the current I in the ring.

(e) Find the force required to hold the dipole in place (not the torque).

From now on, there is no permeable medium. Ignore any radiation loss for all parts.

(f) The dipole (mass M), starts at a distance h from the centre of the ring (kept fixed) and pointed towards the centre of the ring (along its axis), and is projected with a small velocity v_0 towards the centre. Find its speed v as a function of h. Ignore gravity.

(g) Consider another scenario, in which the dipole is placed on the axis of a thin infinite magnetic tube with surface conductivity σ (defined as the ratio of surface current density and the electric field) and radius R, placed at an arbitrary location inside it. (You may neglect the self inductance of the solenoid for the sake of this part.) We find that the motion of the dipole in this case is damped. Find the damping parameter of this motion. (Damping parameter is defined as the ratio of the resistive force to the speed.) Ignore gravity.

(h) Determine the terminal velocity of the magnet, assuming that it now falls under gravity. The tube may be considered infinitely long for all calculation purposes in this part.

3 Thermodynamics

3.1 Gaz İçinde Titreşim

İçi gaz dolu kapalı bir ortamda, k yay sabitli bir yaya bağlı m kütleli ve S yüzey alanlı bir plaka titreşim yapmaktadır. Bu titreşimin her periyodunu tamamladıktan sonra genliği $(A' - A)/A = \xi$ kadar azalıyor. Ortamın hacmi ve ortamdaki mol sayısı biliniyorsa, ortamın sıcaklığını ve basıncını hesaplayın.

(Öykü Abla, 2024 Şubat Kampı)

3.2 Sonsuz Çevrim

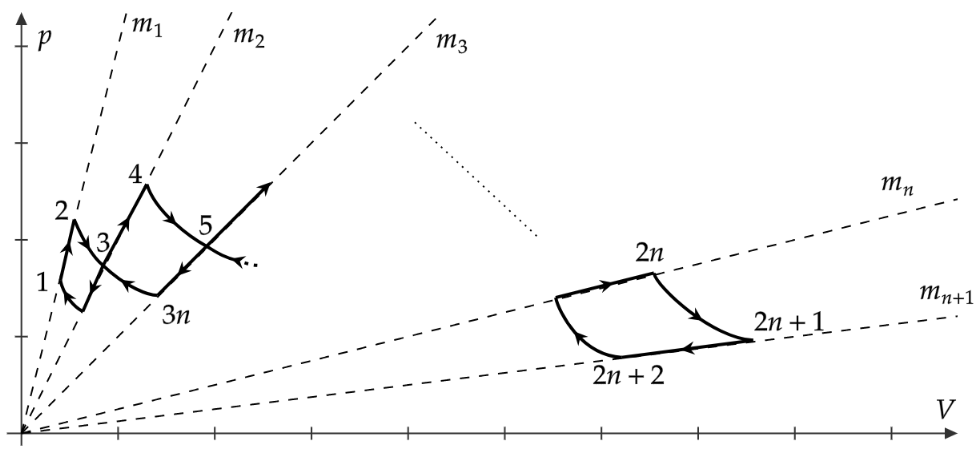
Aşağıdaki şekilde verilen çevrim, sabit eğimli doğrular üzerindeki noktalar ve bu noktaları bağlayan izotermal eğriler üzerinde ilerlemektedir. 1. noktanın sıcaklığı T_0, 2. noktanın sıcaklığı $T_0 + \Delta T$, 4. noktanın sıcaklığı $T_0 + 2\Delta T$ olarak veriliyor. Yani, herhangi bir i için $T_{2i+2} - T_{2i} = \Delta T$ olduğu biliniyor. (2024, 4. Aşama)

a) Doğrusal proseslerin ısı sabiti c nedir?

b) Bu çevrimin ilk iki döngüsünü ($n = 2$) $p - T$ ve $V - T$ grafikleri üstünde gösterin.

c) Çevrimin verimin η'yı n'den bağlı olarak hesaplayın.

d) Çevrimin maksimum ve minimum sıcaklıkları arasında çalışan Carnot makinesinin verimini (η_c) hesaplayın ve $\eta < \eta_c$ olduğunu gösterin.



3.3 Infinite Cycles (Kutay's Version)

Üstteki soruda yaptığınız her şeyi, doğrular yerine denklemi

$$\frac{V}{V_0} = n\left(\frac{p}{p_0}\right)^2$$

şeklinde verilen paraboller varmış gibi tekrar yapın (ki pratik olsun). (Kutay - Kendi çözmedi.)